

Agentic AI for Conversation Analysis

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Abstract—Call center recordings represent a valuable yet underutilized source of business intelligence, containing insights into customer preferences, service effectiveness, and agent performance. However, traditional methods of manual review are resource-intensive, subjective, and poorly scalable across large organizations. To address this challenge, we propose **Agentic AI** — a modular framework that leverages agent-oriented design to automate conversation analysis in call centers. The system processes audio recordings end-to-end: downloading files from remote sources, transcribing speech to text using state-of-the-art models, summarizing conversations via large language models (LLMs), and presenting structured metadata through an interactive dashboard. **Agentic AI** is powered by Whisper for robust speech-to-text transcription, HuggingFace transformers for semantic summarization, and CrewAI for orchestrating agent-task-tool interactions. The architecture divides responsibilities across four dedicated agents: audio acquisition, transcription, summarization, and dashboard visualization. Each agent operates independently and communicates via structured task outputs, enabling modular updates and fine-tuned control across stages. This design eliminates bottlenecks common in linear pipelines and simplifies integration with external APIs and databases. The output includes download-ready transcripts, compressed summaries, and metadata such as agent ID, call duration, and conversation type. Results are displayed through a user-centric interface built with Dash, allowing stakeholders to track call quality, detect patterns across agents, and make data-driven decisions. The system has processed dozens of real-world sales calls, successfully extracting context-rich narratives that enhance sales planning and customer engagement strategy. **Agentic AI** combines speech processing, language modeling, and agentic orchestration into a scalable framework for call intelligence. It transforms passive audio logs into actionable insights, reducing manual workload while improving the consistency and depth of analysis — paving the way for smarter call centers and responsive sales teams.

Index Terms—Agentic AI, CrewAI, Speech-to-Text, Whisper, HuggingFace, Dash, Large Language Models, Call Analytics, Conversation Intelligence

I. INTRODUCTION

Call centers generate vast amounts of data through interactions between customers and agents. While these conversations are rich in insights, analyzing hundreds or thousands of hours of audio manually is labor-intensive, error-prone, and slow. In an enterprise setting, where real-time data and timely feedback are crucial for improving service and optimizing performance, traditional manual review is simply not scalable.

This paper introduces **Agentic AI**, an intelligent system that automates the entire process of conversation analysis from audio acquisition to actionable business insights. By using **large**

language models (LLMs), **speech-to-text technologies**, and a modular agent architecture, **Agentic AI** is designed to process call center conversations efficiently, generate summaries, and present the data in an easy-to-digest, visual format through an interactive dashboard. In this paper, we define key components of the system, discuss the process, and provide an evaluation of its effectiveness.

II. MOTIVATION AND PROBLEM STATEMENT

In conventional call centers, human agents are tasked with manually reviewing call recordings to evaluate customer satisfaction, analyze agent performance, or extract specific topics of interest (e.g., insurance inquiries, customer complaints). However, this approach faces several limitations:

- **Time-consuming and inconsistent:** Manual review is labor-intensive, prone to human error, and inconsistent in its evaluation standards.
- **Scalability issues:** Given the sheer volume of data generated, this method cannot scale effectively to handle large volumes of calls.
- **Lack of real-time feedback:** Agents or supervisors typically receive delayed feedback, which impacts their ability to make real-time adjustments during interactions.
- **Limited insights:** Manual evaluation often focuses on a few specific metrics (e.g., call length, resolution) and overlooks broader trends or patterns in conversations.

Agentic AI aims to address these problems by automating the entire call analysis workflow, leveraging **modular agents** that perform specific tasks and combine their results into a comprehensive analysis.

III. BACKGROUND AND TOOLS

Agentic AI relies on several open-source tools and frameworks to build its analysis pipeline:

- **FFmpeg:** An open-source multimedia framework used for processing audio files. It can convert between different audio formats (e.g., MP3 to WAV), perform noise reduction, and extract features needed for transcription.
- **Whisper:** Whisper is an open-source **speech-to-text** model developed by OpenAI [1].
- **HuggingFace Transformers:** HuggingFace provides a suite of pre-trained **large language models (LLMs)** that can be applied to text summarization, sentiment analysis, and entity recognition. We leverage these models to

summarize call transcripts and extract key information, such as customer intent and resolution [2].

- **CrewAI:** CrewAI is a framework that allows the orchestration of various AI tools and tasks through modular agents. Each agent in the pipeline performs a specific task, such as audio processing, transcription, summarization, or presentation of the results [3].
- **Dash:** Dash is a Python framework for building interactive, web-based dashboards. It allows us to present the results of the conversation analysis in an intuitive format, including metadata (e.g., agent names, call types) and summaries generated by HuggingFace models [4].

These tools collectively form the **Agentic AI pipeline**, where audio is processed, transcribed, summarized, and then presented in a dashboard. Below, we provide an overview of the modular agents involved in the process.

IV. THEORY AND METHODOLOGY

A. Agentic Workflow

The core of Agentic AI is its **modular agent-based architecture**. In an agentic workflow, each agent is a self-contained unit that performs a specific task and communicates with other agents to exchange data. This architecture ensures that the system is **scalable**, **flexible**, and **extensible**.

The main agents in our system are:

- **AudioAgent:** This agent is responsible for downloading the call recordings from the metadata provided by the Convoso API. It retrieves the audio files in their original format and stores them locally.
- **TranscribeAgent:** This agent handles the speech-to-text conversion. It uses the Whisper model to transcribe the audio recordings into text, enabling further analysis.
- **SummarizeAgent:** This agent applies HuggingFace’s pre-trained language models to the transcripts to generate **summaries**. These summaries are designed to highlight key information, such as customer concerns, the outcome of the call, and any relevant action items.
- **DashboardAgent:** This agent is responsible for displaying the results of the analysis in an interactive **Dash** dashboard. The dashboard visualizes the metadata (e.g., agent ID, call duration, topics discussed) and provides options to filter or search for specific calls. It also presents the call summaries and allows users to download the relevant files.

B. Data Flow

The data flow in the Agentic AI system begins with the **metadata acquisition** from the Convoso API. This metadata includes call details such as the **agent ID**, **call duration**, **call type**, and **call ID**. This data is essential for organizing the conversation logs and presenting the information in a structured format.

Once the metadata is obtained, the **AudioAgent** downloads the corresponding audio files. These files are then processed by the **TranscribeAgent**, which converts the spoken content

into text. The resulting transcripts are fed into the **SummarizeAgent**, which uses pre-trained HuggingFace models to generate concise and informative summaries.

Finally, the **DashboardAgent** presents these insights in an interactive format, allowing users to search, filter, and view summaries, as well as download the necessary files (audio, transcript, and summary).

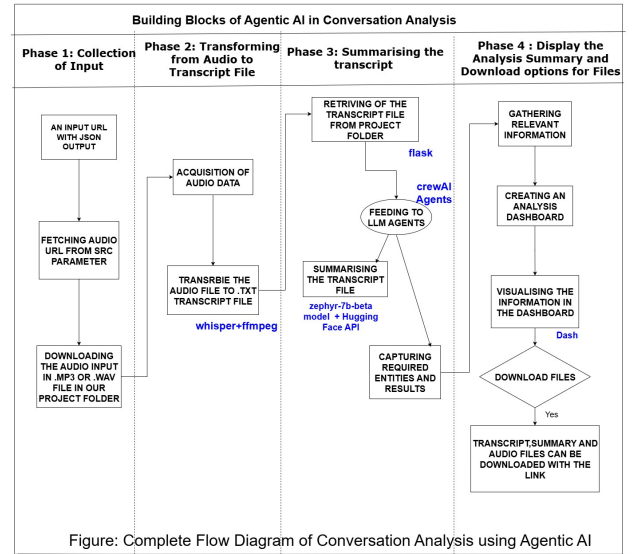
C. Language Models for Summarization

Summarization is a key task in the Agentic AI pipeline. Large language models (LLMs) such as GPT-3 or BERT-based models are used to generate summaries. These models are pre-trained on vast amounts of text data and can capture the essence of a conversation by identifying important themes and filtering out irrelevant content. The summarization process is as follows:

- **Text extraction:** After transcription, the transcript is processed by the summarization model.
- **Contextual summarization:** The LLM analyzes the transcript, extracting the core content (e.g., customer complaints, product inquiries, resolutions).
- **Summary generation:** A concise summary is generated that provides actionable insights, allowing users to quickly understand the context and outcome of the call.

This automated summarization ensures that no important details are overlooked and saves time by reducing the need for manual review.

V. SYSTEM FLOWCHART AND DASHBOARD



The flowchart above illustrates the data flow within the system, from metadata acquisition to final summary presentation. Each step is performed by a dedicated agent, and their outputs are passed on to subsequent agents for further processing.

The dashboard shown in Figure 1 presents the results in an intuitive and user-friendly format. Users can filter the data by call type, agent, or other relevant parameters, providing

From an engineering perspective, the use of **CrewAI** reduces the complexity of orchestrating tasks and agents, simplifying debugging, and enabling quick expansion of the system to include new agents or workflows.

VIII. FUTURE WORK

We plan to enhance Agentic AI with the following features:

- **Real-time transcription for live feedback:** Allow supervisors to monitor calls as they happen and provide real-time coaching.
- **Sentiment analysis:** Detect customer sentiment and classify conversations as positive, negative, or neutral.
- **Intent recognition:** Understand customer intent to route calls to the appropriate department or agent.
- **Multimodal integration:** Incorporate audio, text, and visual cues for a richer analysis of customer interactions.
- **Reinforcement learning:** Adjust agent behavior based on performance feedback.

IX. CONCLUSION

Agentic AI transforms conversation logs into strategic business intelligence. By automating the transcription, summarization, and analysis of call recordings, it empowers organizations to optimize sales and customer service operations. With modular agents, real-time processing, and actionable insights, Agentic AI streamlines the call center evaluation process, making it scalable and efficient.

REFERENCES

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The image shows a web interface titled "Input URL for Summarization". It features a text input field containing a long URL: "https://api.convoso.com/v1/log/retrieve?auth_token=3gq83e2g8pa0hg8rva0q8o6im3vg&id=93912843&le". To the right of the input field are two buttons: "Submit" (green) and "Reset" (red). Below the input field, there is a red error message that says "Please enter a valid URL."

Fig. 1. Input Dashboard UI Requesting for URL

deep insights into the calls and enabling decision-makers to act promptly.

VI. RESULTS

Using real-world call recordings, the Agentic AI system processed interactions across different domains, such as insurance inquiries and service complaints. The pipeline generated actionable insights such as:

- **Call Duration:** 16 to 46 minutes.
- **Topics:** Insurance inquiry, health disclosures, quote negotiations.
- **Metadata:** Presented in a structured format, including call type, date, and duration.

These results were visualized on the dashboard.

The image shows a dashboard titled "Call Summarizer". It has a header with a phone icon and the title. Below the header is a text input field with the same URL as in Fig. 1, and a "Submit" button. The dashboard is divided into two main sections: "Metadata" and "Summary". The "Metadata" section lists several details: Customer Name: Vincent Mickens, Agent Name: Deanna Nunn, Duration: 16 min 24 sec, Call Status: Qualified Life Only Front, Type of Issue: INBOUND, and Customer Satisfaction: 5 stars. The "Summary" section contains a paragraph: "The customer Vincent Mickens spoke with agent Deanna Nunn in an inbound call. The issue was addressed and the customer seemed satisfied. The customer seemed happy with the call and the agent was able to get the issue back on the phone. The call was recorded and returned to the customer's satisfaction." At the bottom of the dashboard are three buttons: "Download Transcript", "Download Summary", and "Download Audio".

Fig. 2. Final Dashboard After Submitting the Input URL

VII. DISCUSSION

Agentic AI provides several advantages over traditional manual review:

- **Efficiency:** Automates the transcription, summarization, and analysis of call recordings, reducing the workload on human agents.
- **Scalability:** The modular nature of the agents allows the system to scale with increasing call volumes.
- **Real-time Feedback:** The system can be extended to provide real-time transcription and summary generation, which can help agents receive immediate feedback.